

Al meet PLC Schneider

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Config awal

The screenshot shows the 'EcoTruxure Machine Expert - Basic' software interface. The top bar includes the logo, project name 'tcp', and status indicators 'No error' and 'Not Connected'. Below this is a toolbar with various icons. The main window has a green header with tabs: 'Properties', 'Configuration', 'Programming', 'Display', and 'Commissioning'. The 'Configuration' tab is active, showing a left sidebar with a tree view containing 'Project Properties', 'Front Page', 'Company', 'Project Information', 'Project Protection', and 'Application Protection' (which is selected and highlighted). The main area displays the 'Application Protection' settings. A yellow warning icon and text state: 'You should set active both application protections'. There are two sections: 'Read protection' and 'Write protection'. Each section has radio buttons for 'Active' and 'Inactive'. The 'Inactive' option is selected for both. For 'Read protection', the 'Inactive' description is 'Uploading the application from the controller is unrestricted'. For 'Write protection', the 'Inactive' description is 'Downloading an application to the controller or modifying the application in the controller is unrestricted'. Each section also has input fields for 'Password' and 'Confirmation'.

EcoTruxure
Machine Expert - Basic tcp

No error Not Connected

Properties Configuration Programming Display Commissioning

Project Properties
Front Page
Company
Project Information
Project Protection
Application Protection

Application Protection

You should set active both application protections

Read protection

☐ Active

Password

Confirmation

☒ Inactive Uploading the application from the controller is unrestricted

Write protection

☐ Active

Password

Confirmation

☒ Inactive Downloading an application to the controller or modifying the application in the controller is unrestricted

Pilih PLC TM221CE16R

Messages

MyController (TM221CE16R)



Device information

TM221CE16R



Messages

Device description

TM221CE16R (screw)
9 digital inputs, 7 relay outputs (2 A), 2 analog inputs, 1 serial line port, 1 Ethernet port, 100-240 Vac power supply controller with removable terminal blocks.

M221 Logic Controllers

Refere...	Power supply	Comm. Ports
TM221M32TK	24 Vdc	2 SL
TM221M16T/G	24 Vdc	2 SL
TM221M16R/G	24 Vdc	2 SL
TM221CE40U	24 Vdc	1 SL + 1 ETH
TM221CE40T	24 Vdc	1 SL + 1 ETH
TM221CE40R	100...240 Vac	1 SL + 1 ETH
TM221CE24U	24 Vdc	1 SL + 1 ETH
TM221CE24T	24 Vdc	1 SL + 1 ETH
TM221CE24R	100...240 Vac	1 SL + 1 ETH
TM221CE16U	24 Vdc	1 SL + 1 ETH
TM221CE16T	24 Vdc	1 SL + 1 ETH
TM221CE16R	100...240 Vac	1 SL + 1 ETH
TM221C40U	24 Vdc	1 SL
TM221C40T	24 Vdc	1 SL
TM221C40R	100...240 Vac	1 SL
TM221C24U	24 Vdc	1 SL

> TM3 Digital I/O Modules

> TM3 Analog I/O Modules

> TM2 Digital I/O Modules

> TM2 Analog I/O Modules

> TM3 Expert I/O Modules

> M221 Cartridges

Device description

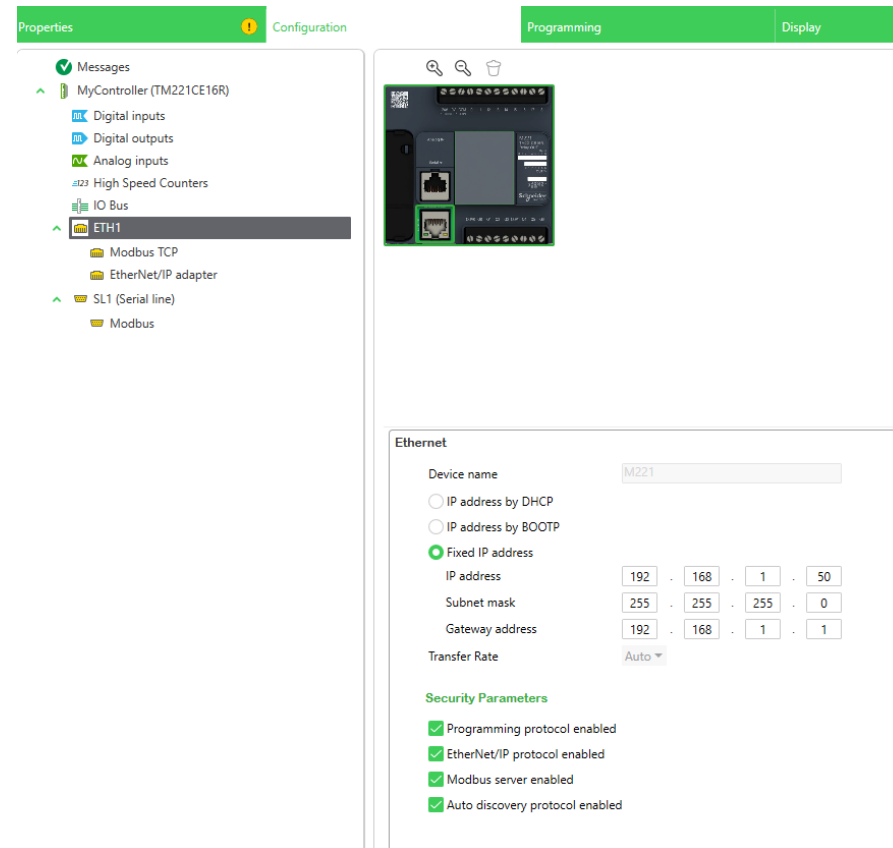
TM221CE16R (screw)
9 digital inputs, 7 relay outputs (2 A), 2 analog inputs, 1 serial line port, 1 Ethernet port, 100-240 Vac power supply controller with removable terminal blocks.

Power supplied to the IO bus

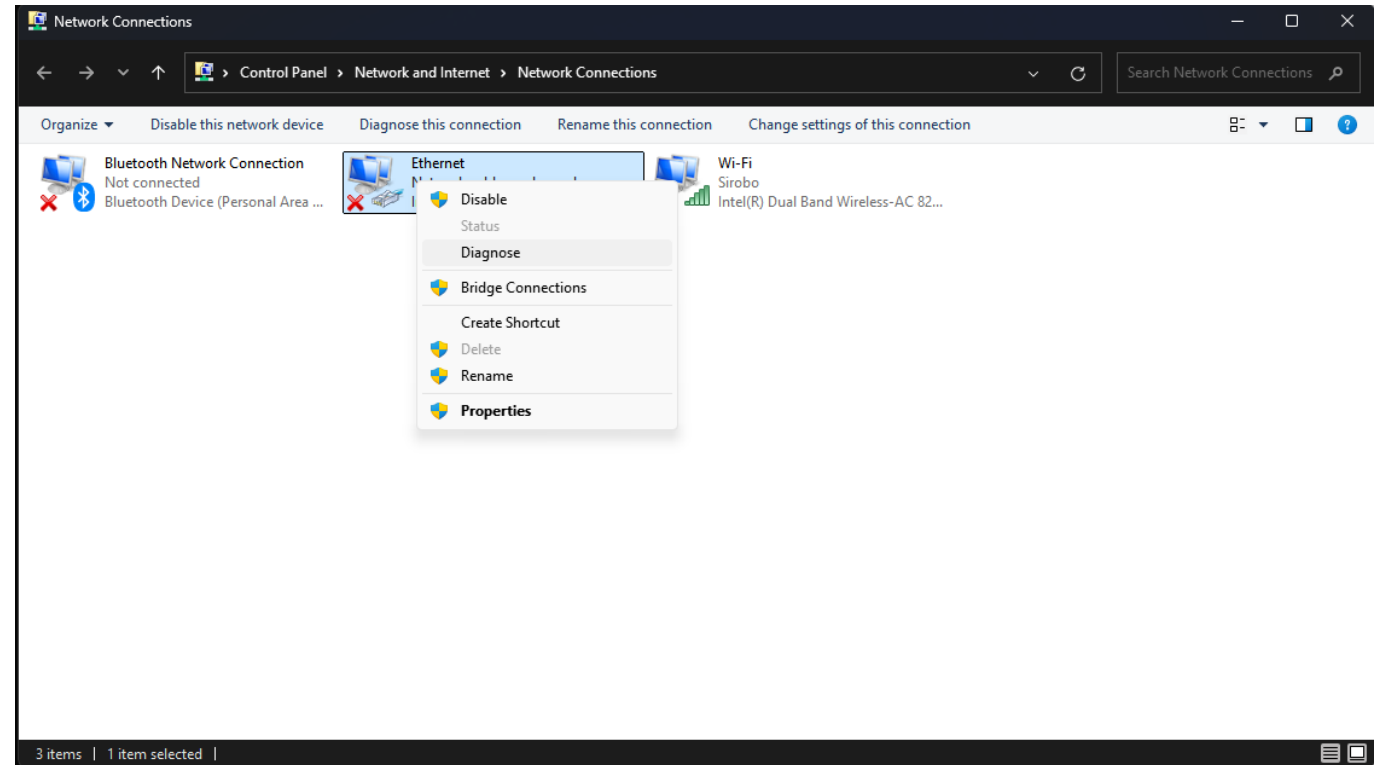
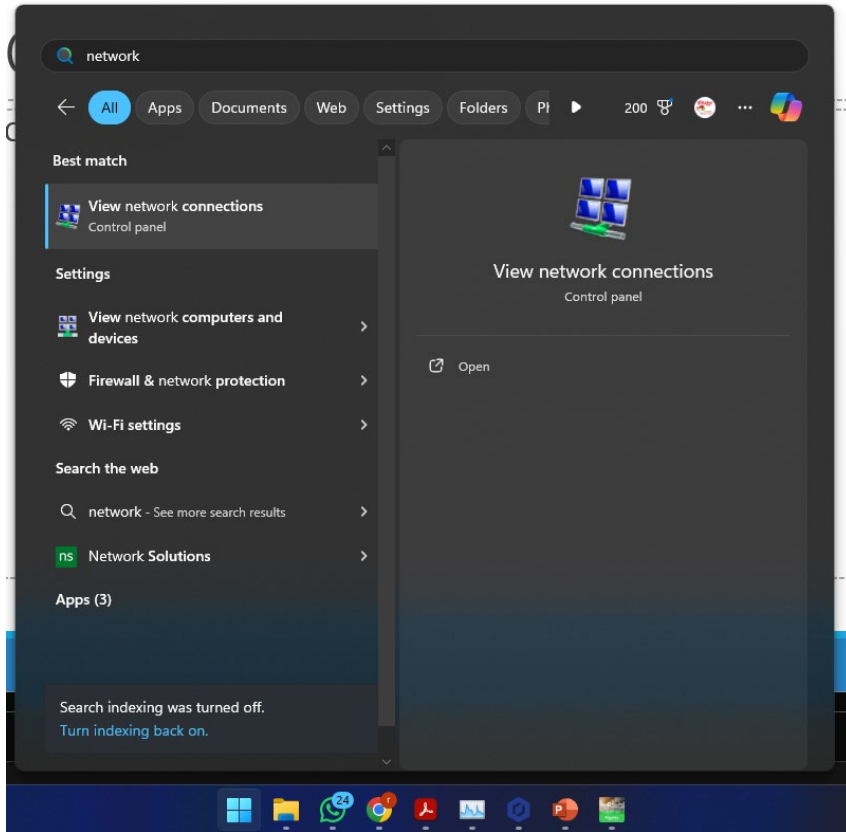
5V	24V
325 mA	120 mA



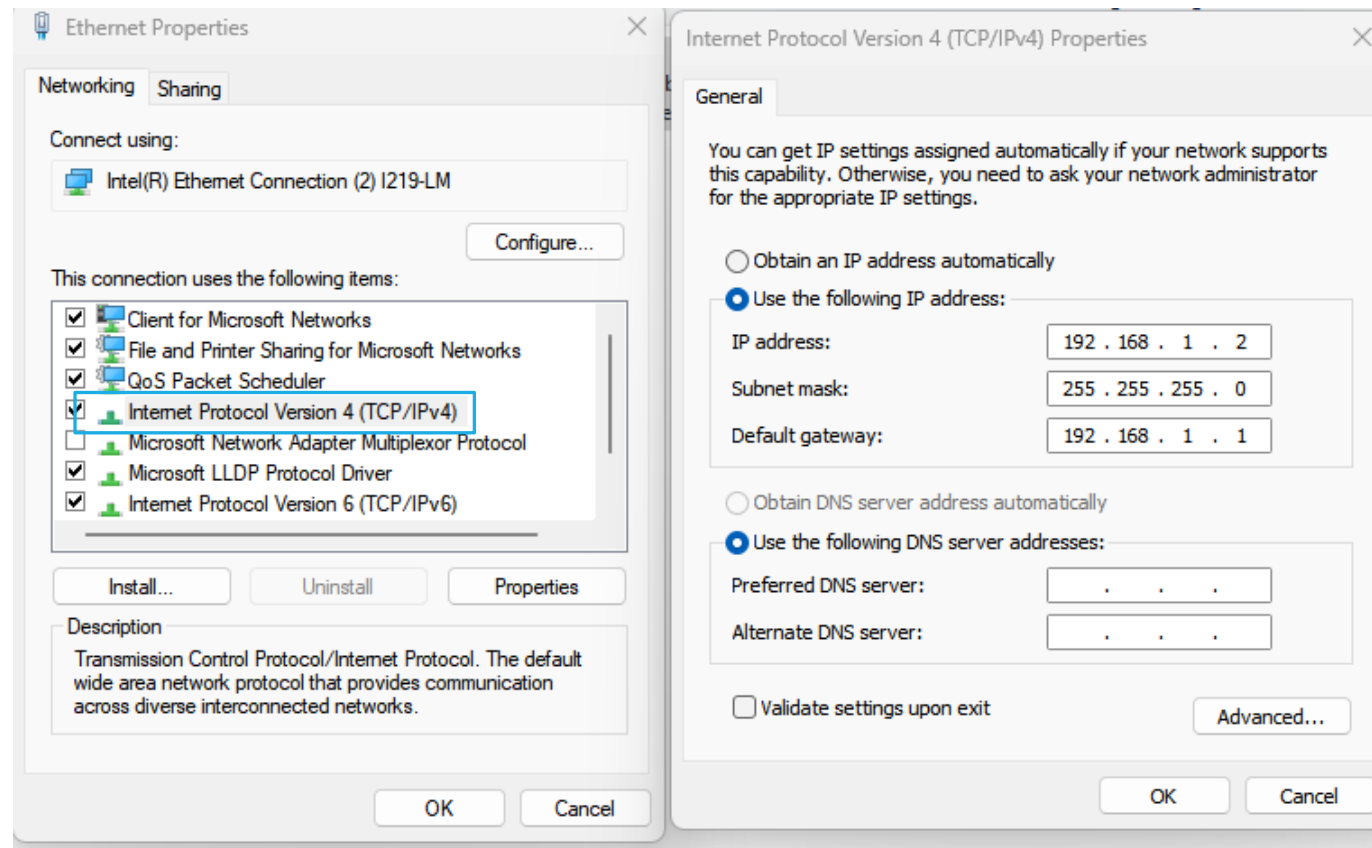
Setting Alamat IP di PLC 192.168.1.50



Setting Alamat IP di PC



Setting Alamat IP di PC 192.168.1.2



Setting Alamat IP dari PLC ke PC 192.168.1.2

Properties Configuration Programming Display Commissioning

Messages

- MyController (TM221CE16R)
 - Digital inputs
 - Digital outputs
 - Analog inputs
 - High Speed Counters
 - IO Bus
 - ETH1
 - Modbus TCP
 - EtherNet/IP adapter
 - SL1 (Serial line)
 - Modbus

Modbus TCP

Modbus mapping

☐ Enabled Unit ID Output registers (%IWM) Input registers (%QWM)

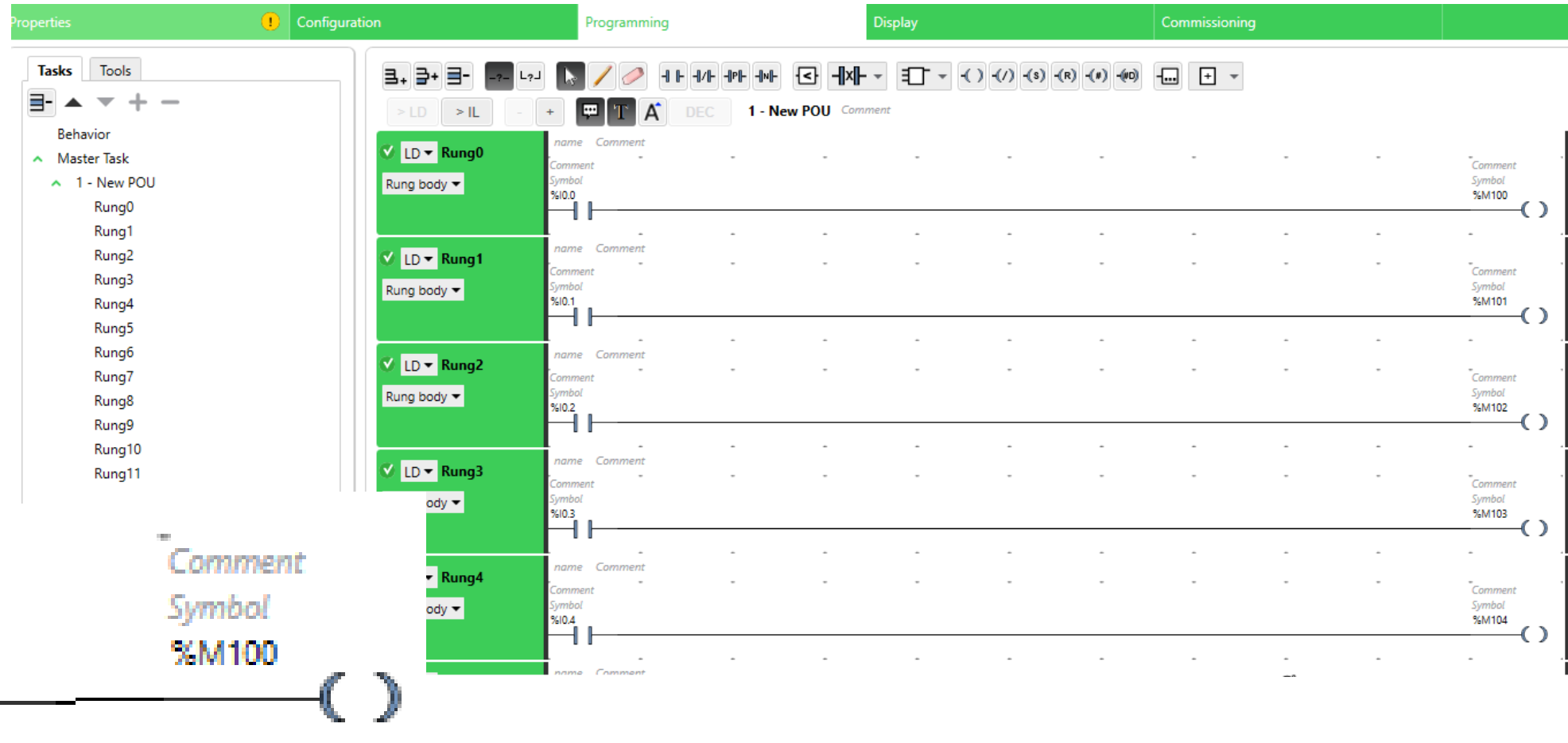
Client mode: remote device table (max 16)

☐ Enable Modbus TCP IOScanner

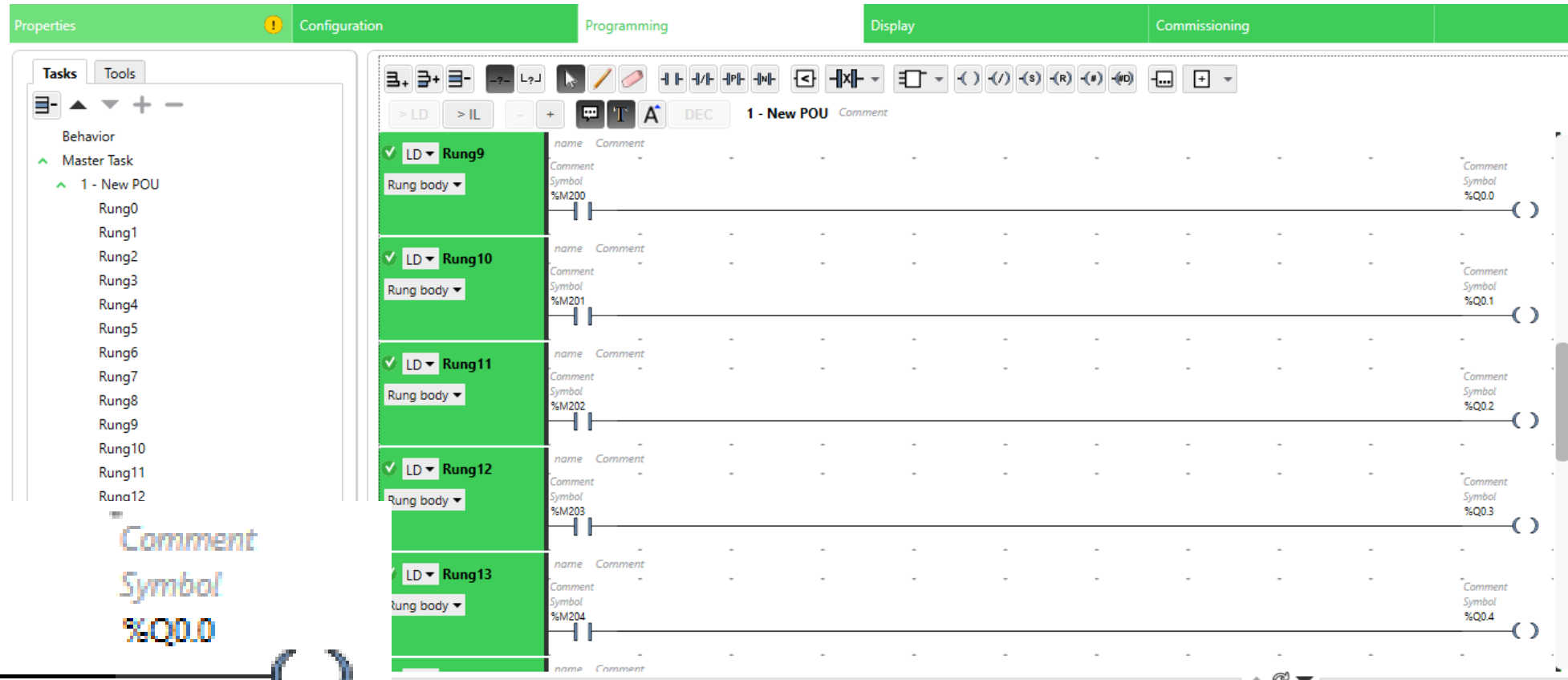
IP address: ☒ Generic ☐ Drive ATV12 ☐ Predefined ATS22 Altistart

ID	Name	Address	Type	Index	IP address	Resp...	Reset variable	Scanned	Init R...	Init. requests	Chan...	Channels
0	Device 0		Generic device	1	192.168.1.2	10			255	...	255	...

Baca I0 - I3 sambungkan ke M100 - M103



Baca M200 - M203 sambungkan ke Q0 - Q03



Function Block Write M100-103

Configuration

Link: 3 - ETH1

Id: 1

Timeout: 100

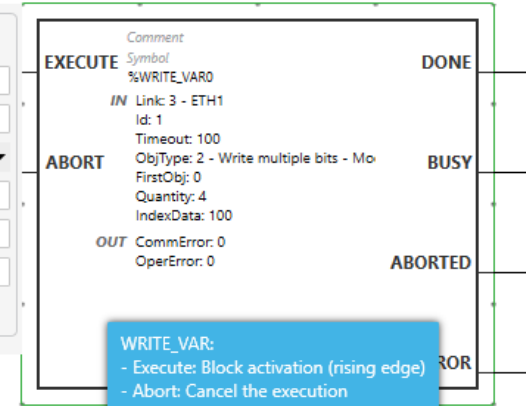
ObjType: 2 - Write multiple bits - Modbus 0x0F

FirstObj: 0

Quantity: 4

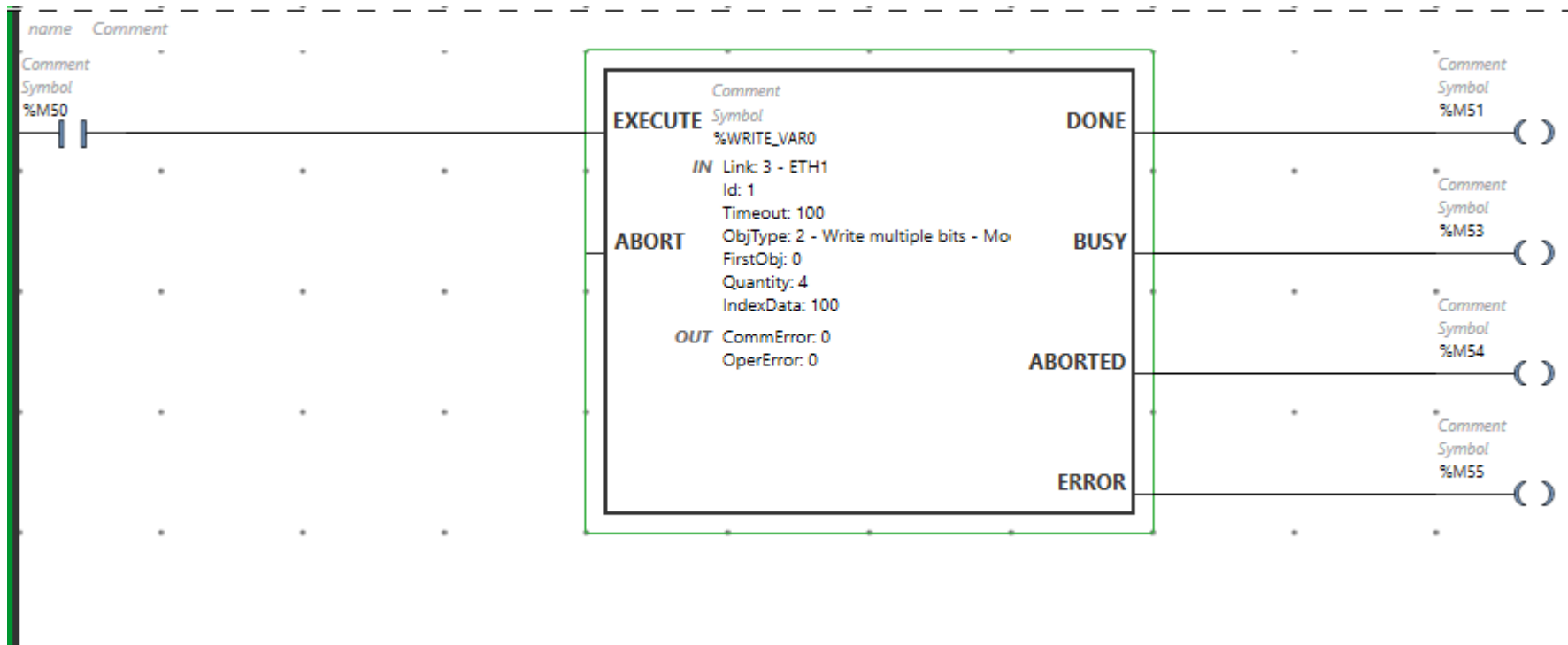
IndexData: 100

Apply Cancel



WRITE_VAR:

- Execute: Block activation (rising edge)
- Abort: Cancel the execution
- Done: Execution ended successfully
- Busy: Execution ongoing
- Aborted: Execution canceled
- Error: Execution ended unsuccessfully



ent

IOIO

123

1123

11123

ABC

E

DRV

UDFB

Read Var

Write Var

Write Var

Send Receive Message

Send Receive SMS

Function Block Read M200-203

Configuration

Link: 3 - ETH1

Id: 1

Timeout: 100

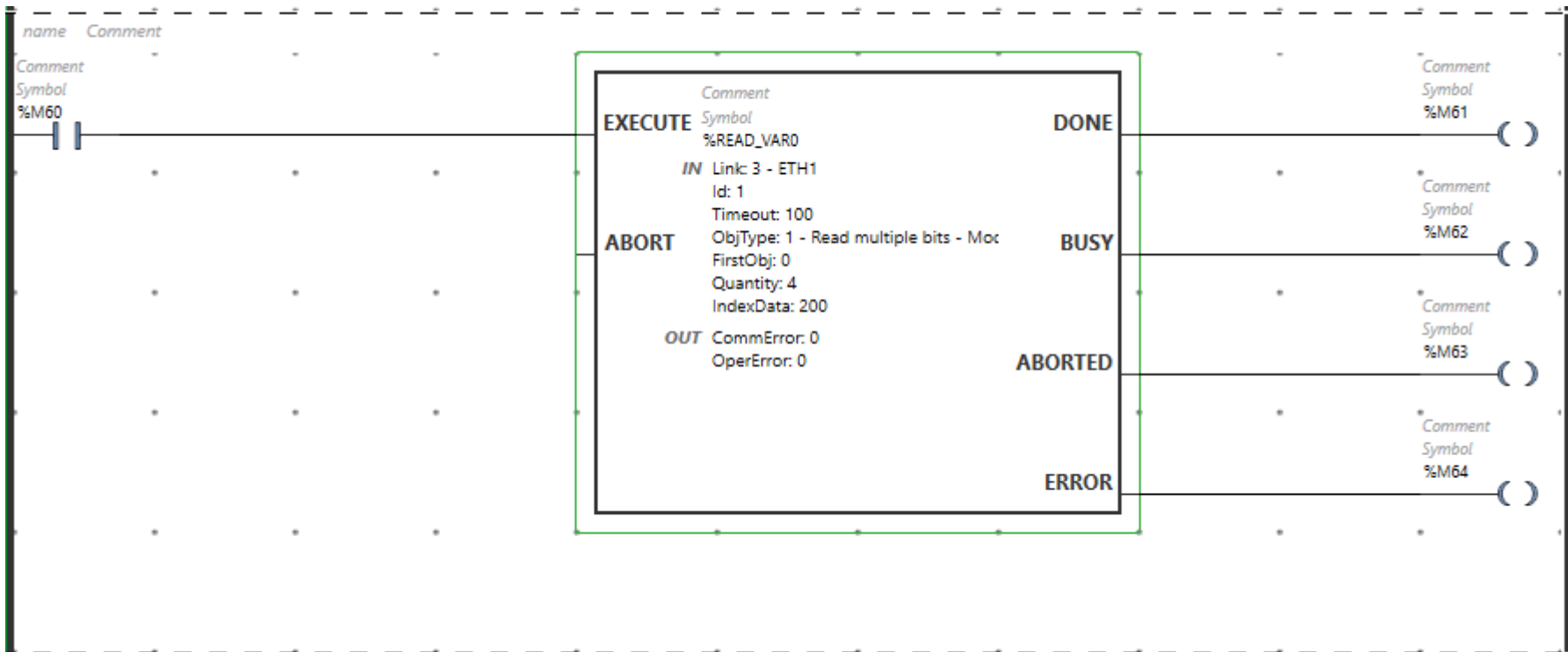
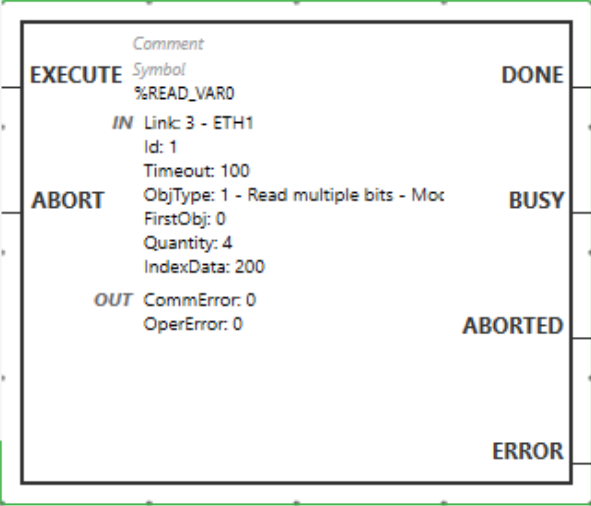
ObjType: 1 - Read multiple bits - Modbus 0x02

FirstObj: 0

Quantity: 4

IndexData: 200

Apply Cancel



Toolbar icons: [Icons for various functions]

Read Var

Write Var

Write Read Var

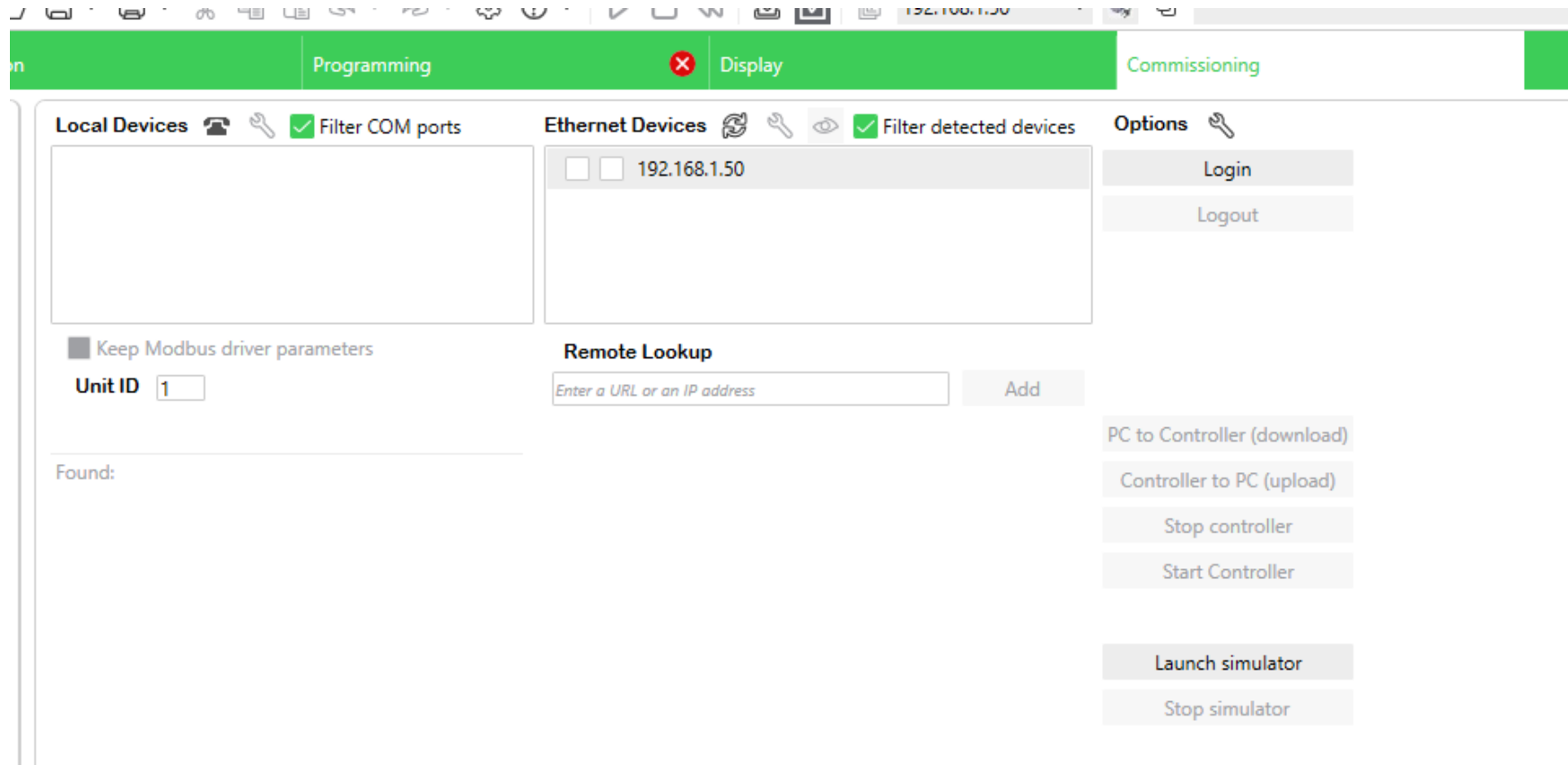
Send Receive Message

Send Receive SMS

Read Var

Upload Program ke PLC via USB

Login - Download - Start



Program Python tes.py untuk koneksi modbus tcp

<https://chatgpt.com/share/68261b22-3b84-800f-9b37-f3d77403f6fa>

Buatkan program python

Connect ke PLC via Modbus TCP

alamat PLC 192.168.1.50

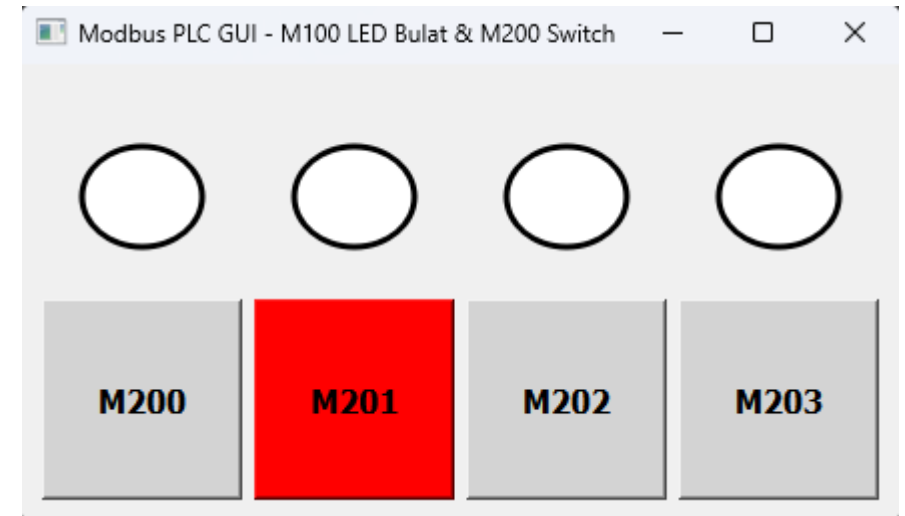
alamat PC 192.168.1.2

baca 4 data M100-M103 buatkan seperti LED

irim 4 data M200-203 buatkan seperti Switch

buatkan GUI dengan QT

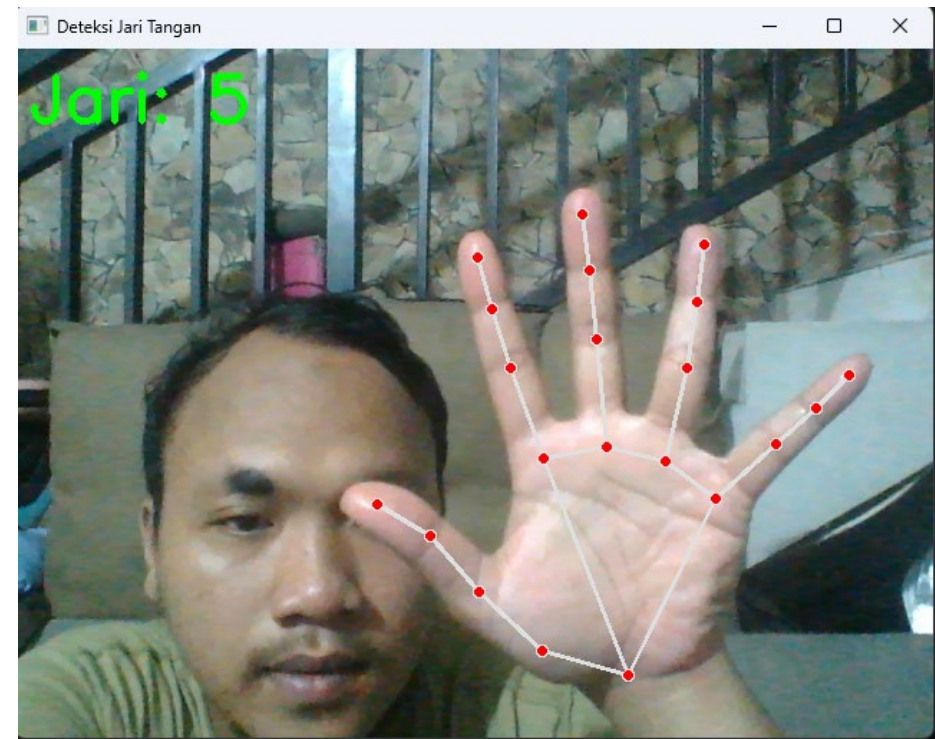
tampilkan LOG data di console



Program Python tes2.py untuk deteksi jari tangan

<https://chatgpt.com/share/68261d76-ee1c-800f-b573-645bda348319>

buatkan program python untuk deteksi jari tangan



Program Python tes3.py untuk gabungkan deteksi jari tangan dan koneksi modbus tcp

<https://chatgpt.com/share/68262721-a978-800f-b4d0-2515e64697b5>

Copy Paste program tes modbus tcp, shift + enter

Copy Paste program deteksi jari, shift + enter

gabungkan kedua program diatas,

ketika jari telunjuk diangkat M200 on

ketika jari tengah diangkat M201 on

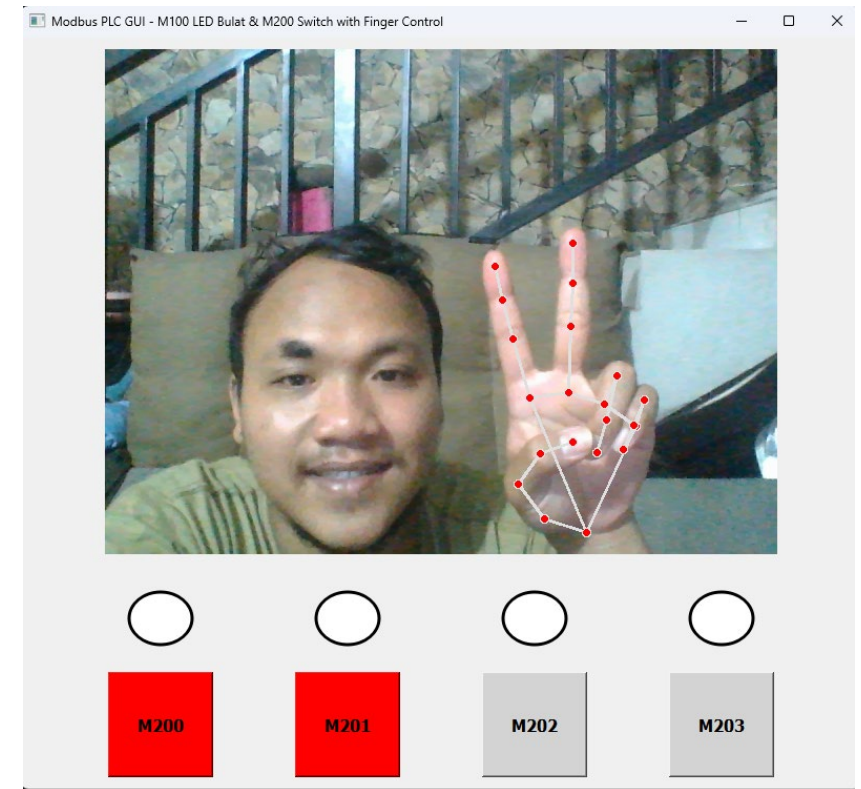
ketika jari manis diangkat M202 on

ketika jari kelingking diangkat M203 on

Ketika mengepal atau turun, akan off

Ketika tidak terdeteksi jari, toggle tetap bisa di on off kan

tampilkan webcam di gui dan bawahnya gui seelumnya



Tugas

Video semua kegiatan ketika mencoba PLC

Buat Video dengan screenrecorder layar dan webcam

Jelaskan program PLC dari config hingga upload, serta config IP PC

Jelaskan program python tes koneksi modbus tcp

Jelaskan program python deteksi jari tangan

Jelaskan program python yang sudah terintergrasi

Upload di youtube kalian